数据库课程实验报告

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1 创建关系表并导入数据

1.1 创建关系表

关系模式:
student(NO,NAME,SEX,BIRTHDAY,CLASS)
teacher(NO,NAME,SEX,BIRTHDAY,DEPART)
course(CNO,CNAME,TNO)
score(NO,CNO,DEGREE)

```
create schema ST;
use ST;
#
create table student
(
NO char(11) primary key,
NAME char(3),
SEX char(6),
BIRTHDAY date,
CLASS smallint
);

create table teacher
(
NO char(7) primary key,
NAME char(3),
SEX char(6),
BIRTHDAY date,
PROF char(19),
DEPART smallint
);

create table course
(
CNO char(8) primary key,
CNAME char(30),
TNO char(7)
);

create table score
```

```
(
NO char(11),  #
CNO char(8),
DEGREE smallint,
primary key(NO,CNO)
);

/*    schema*/
```

Tables_in_st
course
score
student
teacher

schema

Field	Type	Null	Key	Default
NO	char(11)	NO	PRI	NULL
NAME	char(3)	YES		NULL
SEX	char(6)	YES		NULL
BIRTHDAY	date	YES		NULL
CLASS	smallint(6)	YES		NULL

(b)student

Field	Type	Null	Key	Default
NO	char(7)	NO	PRI	NULL
NAME	char(3)	YES		NULL
SEX	char(6)	YES		NULL
BIRTHDAY	date	YES		NULL
PROF	char(19)	YES		NULL
DEPART	smallint(6)	YES		NULL

(c)teacher

Field	Type	Null	Key	Default
CNO	char(8)	NO	PRI	NULL
CNAME	char(30)	YES		NULL
TNO	char(7)	YES		NULL

(d)course

Field	Type	Null	Key	Default
NO	char(11)	NO	PRI	NULL
CNO	char(8)	NO	PRI	NULL
DEGREE	smallint(6)	YES		NULL

(b)score

图 1: schema

1.2 导入数据

导入数据后查询每个表的数据

```
SELECT * FROM ST.course;
SELECT * FROM ST.score;
SELECT * FROM ST.student;
SELECT * FROM ST.teacher;
```

CNO	CNAME	TNO
20181009	Convex_Optimization	TA80023
20181102	Database	TA80021
20181103	Machine_Learning	TA80023
20181104	Operating_System	TA80025
20181105	Natural_Language_Processing	TA80027
20181106	Artificial_Intelligence	TA80029
20181107	Comprehensive_English	TA80028
20181108	Signal_Control	TA80022
20181109	Computer_Network	TA80024
20181110	Pattern_Recognition	TA80022
NULL	NULL	NULL

(a)course data

NO	CNO	DEGREE
SA170110001	20181102	89
SA170110001	20181103	72
SA170110001	20181107	89
SA170110001	20181110	91
SA170110002	20181102	94
SA170110002	20181103	81
SA170110002	20181110	95
SA170110003	20181102	89
SA170110003	20181103	92
SA170110003	20181110	82
SA170110004	20181102	95
SA170110004	20181103	68
SA170110005	20181009	76
SA170110005	20181102	93
SA170110005	20181103	71
SA170110005	20181107	74
SA170110006	20181009	91
SA170110006	20181102	76
SA170110006	20181103	92
SA170110007	20181009	82
SA170110007	20181102	79

(b)score data 1

SA170110008	20181009	91
SA170110008	20181102	82
SA170110008	20181107	79
SA170110008	20181110	69
SA170110010	20181104	81
SA170110011	20181104	84
SA170110011	20181106	75
SA170110012	20181104	78
SA170110013	20181104	89
SA170110014	20181106	83
SA170110015	20181105	74
SA170110015	20181107	91
SA170110016	20181106	77
SA170110017	20181105	80
SA170110018	20181105	92
SA170110018	20181107	84
SA170110019	20181105	96
SA170110019	20181107	88
SA170110020	20181105	85
SA170110020	20181107	86
NULL	NULL	NULL

(c)score data 2

NO	NAME	SEX	BIRTHDAY	CLASS
SA170110001	LY	male	1994-12-08	1
SA170110002	WKS	female	1995-09-12	2
SA170110003	WPY	male	1994-04-29	1
SA170110004	HTN	male	1995-03-15	2
SA170110005	ZDK	female	1994-08-12	1
SA170110006	QDS	male	1994-06-25	1
SA170110007	PJD	male	1994-06-14	2
SA170110008	LA	male	1994-08-23	3
SA170110009	UN	female	1994-06-23	3
SA170110010	WX	male	1995-02-24	2
SA170110011	QY	female	1994-05-08	4
SA170110012	ZZY	male	1993-06-26	5
SA170110013	WWT	male	1995-11-17	4
SA170110014	LSS	male	1993-01-28	1
SA170110015	XH	female	1996-10-09	2
SA170110016	LC	female	1995-11-30	1
SA170110017	TX	male	1996-05-16	2
SA170110018	ZPL	male	1995-12-02	4
SA170110019	WFL	female	1996-02-13	5
SA170110020	XY	female	1993-02-14	1
NULL	NULL	NULL	NULL	NULL

(d)student data

SA170110010	20181104	81
SA170110011	20181104	84
SA170110011	20181106	75
SA170110012	20181104	78
SA170110013	20181104	89
SA170110014	20181106	83
SA170110015	20181105	74
SA170110015	20181107	91
SA170110016	20181106	77
SA170110017	20181105	80
SA170110018	20181105	92
SA170110018	20181107	84
SA170110019	20181105	96
SA170110019	20181107	88
SA170110020	20181105	85
SA170110020	20181107	86
NULL	NULL	NULL

(b)teacher data

图 2: data import

2 SQL 语句

2.1 修改基本表

1. 假设所有学生都是 11 系学生，在学生表 student 中增加一个新的属性列 DEPART(系)，类型为 char(20);

```sql
alter table student add DEPART char(20) default '11';
SHOW COLUMNS FROM student;
```

Field	Type	Null	Key	Default	Extra
NO	char(11)	NO	PRI	NULL	
NAME	char(3)	YES		NULL	
SEX	char(6)	YES		NULL	
BIRTHDAY	date	YES		NULL	
CLASS	smallint(6)	YES		NULL	
DEPART	char(20)	YES		11	

图 3: problem 1

2. 将上述新增的 DEPART 属性的数据类型修改为 int

```sql
/*alter dose not work in sql*/
alter table student modify column DEPART int default 11;
SHOW COLUMNS FROM student;
```

Field	Type	Null	Key	Default	Extra
NO	char(11)	NO	PRI	NULL	
NAME	char(3)	YES		NULL	
SEX	char(6)	YES		NULL	
BIRTHDAY	date	YES		NULL	
CLASS	smallint(6)	YES		NULL	
DEPART	int(11)	YES		11	

图 4: problem 2

3. 更改每个学生的系 (DEOART) 与班级 (CLASS) 相同 hint: 可能需要关闭安全更新模式

```sql
/*                    */
set SQL_SAFE_UPDATES = 0;
update student set DEPART=CLASS;
```

NO	NAME	SEX	BIRTHDAY	CLASS	DEPART
SA170110001	LY	male	1994-12-08	1	1
SA170110002	WKS	female	1995-09-12	2	2
SA170110003	WPY	male	1994-04-29	1	1
SA170110004	HTN	male	1995-03-15	2	2
SA170110005	ZDK	female	1994-08-12	1	1
SA170110006	QDS	male	1994-06-25	1	1
SA170110007	PJD	male	1994-06-14	2	2
SA170110008	LA	male	1994-08-23	3	3
SA170110009	UN	female	1994-06-23	3	3
SA170110010	WX	male	1995-02-24	2	2
SA170110011	QY	female	1994-05-08	4	4
SA170110012	ZZY	male	1993-06-26	5	5
SA170110013	WWT	male	1995-11-17	4	4
SA170110014	LSS	male	1993-01-28	1	1
SA170110015	XH	female	1996-10-09	2	2
SA170110016	LC	female	1995-11-30	1	1
SA170110017	TX	male	1996-05-16	2	2
SA170110018	ZPL	male	1995-12-02	4	4
SA170110019	WFL	female	1996-02-13	5	5
SA170110020	XY	female	1993-02-14	1	1
NULL	NULL	NULL	NULL	NULL	NULL

图 5: problem 3

4. 为 course 表中的字段 TNO 添加外键，并说明外键带来的好处

```
/*                    */
set SQL_SAFE_UPDATES = 0;
update student set DEPART=CLASS;
```

Table: course

Columns:

CNO	char(8) PK
CNAME	char(30)
TNO	char(7)

Related Tables:

Target	teacher (TNO → NO)
On Update	RESTRICT
On Delete	RESTRICT

图 6: problem 4

外键可以保证数据库的参照完整性，保持数据的一致性、完整性，并且方便级联操作

5. 创建一个学生与老师关系表：relationship(SNO,TNO,DEPART)，三个属性分别表示学生号，教师工号，学生所在系别 (类型为 int)

```
/*                    */
create table relationship
(
SNO char(11),
TNO char(7),
DEPART int
);
SHOW COLUMNS FROM relationship;
```

Field	Type	Null	Key	Default	Extra
SNO	char(11)	YES		NULL	
TNO	char(7)	YES		NULL	
DEPART	int(11)	YES		NULL	

图 7: problem 5

6. 为关系表 relationship 添加主键 (SNO,TNO)

```
alter table relationship add primary key(SNO,TNO);
SHOW COLUMNS FROM relationship;
```

Field	Type	Null	Key	Default	Extra
SNO	char(11)	NO	PRI	NULL	
TNO	char(7)	NO	PRI	NULL	
DEPART	int(11)	YES		NULL	

图 8: problem 6

7. 在表 relationship 中插入一条记录，1 系学生"SA170110005" 的导师是"TA80025"

```
insert into relationship values ('SA170110005','TA80025',1);
SELECT * FROM ST.relationship;
```

SNO	TNO	DEPART
SA170110005	TA80025	1
NULL	NULL	NULL

图 9: problem 7

8. 将学生"SA170110005" 的导师改为"TA80021"

```
update relationship set TNO='TA80021' where SNO='SA170110005';
SELECT * FROM ST.relationship;
```

SNO	TNO	DEPART
SA170110005	TA80021	1
NULL	NULL	NULL

图 10: problem 8

9. 若 2 班的学生的导师都是"TA80024"，将这些信息插入到 relationship 表中 (一条指令)

```
insert into relationship(SNO,TNO,DEPART)
select NO,'TA80024',CLASS from student where CLASS=2;
SELECT * FROM ST.relationship;
```

SNO	TNO	DEPART
SA170110002	TA80024	2
SA170110004	TA80024	2
SA170110005	TA80021	1
SA170110007	TA80024	2
SA170110010	TA80024	2
SA170110015	TA80024	2
SA170110017	TA80024	2
NULL	NULL	NULL

图 11: problem 9

2.2 索引

10. 用 create 语句在 course 的 CNO 上建立普通索引 CNO_INDEX

```
create index CNO_INDEX on course(CNO);
/*      SHOW INDEX FROM course;      */
```

Indexes in Table

Visible	Key	Type	Unique	Columns
✓	PRIMARY	BTREE	YES	CNO
✓	TNO	BTREE	NO	TNO
✓	CNO_INDEX	BTREE	NO	CNO

图 12: problem 10

11. 用 create 语句在 student 的学号 NO 上建立唯一索引 NO_INDEX

```
create unique index NO_INDEX on student(NO);
/*      SHOW INDEX FROM student;      */
```

Indexes in Table

Visible	Key	Type	Unique	Columns
✓	PRIMARY	BTREE	YES	NO
✓	NO_INDEX	BTREE	YES	NO

图 13: problem 11

12. 用 create 语句在 score 表上的学号 NO、成绩 DEGREE 上建立复合索引 NODE_INDEX，要求学号为升序，学号相同时成绩为降序

```
create index NODE_INDEX on score(NO ASC,DEGREE DESC);
SHOW INDEX FROM score;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality
score	0	PRIMARY	1	NO	A	19
score	0	PRIMARY	2	CNO	A	41
score	1	NODE_INDEX	1	NO	A	19
▶ score	1	NODE_INDEX	2	DEGREE	D	40

图 14: problem 12

其中 Collation 中 A 表示 ASC，D 表示 DESC

13. 删除 course 上的索引 CNO_INDEX

```
drop index CNO_INDEX on course;
/*      SHOW INDEX FROM course;      */
```

Indexes in Table

Visible	Key	Type	Unique	Columns
✓	PRIMARY	BTREE	YES	CNO
✓	TNO	BTREE	NO	TNO

图 15: problem 13

2.3 查询

14. 查询和"TX" 同性别并同班的所有同学的姓名

```
select NAME from student
where (SEX,CLASS)=(select SEX,CLASS from student where NAME='TX');
```

NAME
HTN
PJD
WX
TX

图 16: problem 14

15. 查询年龄在 25 岁以下的学生姓名和年龄 (只考虑年)

```
/*                                        */
select NAME,year(now())-year(BIRTHDAY) from student
where year(now())-year(BIRTHDAY)<25;
```

NAME	year(now())-year(BIRTHDAY)
WKS	24
HTN	24
WX	24
WWT	24
XH	23
LC	24
TX	23
ZPL	24
WFL	23

图 17: problem 15

16. 查询至少有 2 名女生的班号

```
/*                                        */
select CLASS from student where sex='female' group by CLASS having count(*)>1;
```

CLASS
2
1

图 18: problem 16

17. 查询名字不以"W" 打头的同学记录

```
select * from student where NAME not like 'W%';
```

NO	NAME	SEX	BIRTHDAY	CLASS	DEPART
SA170110001	LY	male	1994-12-08	1	1
SA170110004	HTN	male	1995-03-15	2	2
SA170110005	ZDK	female	1994-08-12	1	1
SA170110006	QDS	male	1994-06-25	1	1
SA170110007	PJD	male	1994-06-14	2	2
SA170110008	LA	male	1994-08-23	3	3
SA170110009	UN	female	1994-06-23	3	3
SA170110011	QY	female	1994-05-08	4	4
SA170110012	ZZY	male	1993-06-26	5	5
SA170110014	LSS	male	1993-01-28	1	1
SA170110015	XH	female	1996-10-09	2	2
SA170110016	LC	female	1995-11-30	1	1
SA170110017	TX	male	1996-05-16	2	2
SA170110018	ZPL	male	1995-12-02	4	4
SA170110020	XY	female	1993-02-14	1	1
NULL	NULL	NULL	NULL	NULL	NULL

图 19: problem 17

18. 查询学生表的全部记录并按班号和年龄从大到小的顺序

```
select * from student order by CLASS,BIRTHDAY;
```

NO	NAME	SEX	BIRTHDAY	CLASS	DEPART
SA170110014	LSS	male	1993-01-28	1	1
SA170110020	XY	female	1993-02-14	1	1
SA170110003	WPY	male	1994-04-29	1	1
SA170110006	QDS	male	1994-06-25	1	1
SA170110005	ZDK	female	1994-08-12	1	1
SA170110001	LY	male	1994-12-08	1	1
SA170110016	LC	female	1995-11-30	1	1
SA170110007	PJD	male	1994-06-14	2	2
SA170110010	WX	male	1995-02-24	2	2
SA170110004	HTN	male	1995-03-15	2	2
SA170110002	WKS	female	1995-09-12	2	2
SA170110017	TX	male	1996-05-16	2	2
SA170110015	XH	female	1996-10-09	2	2
SA170110009	UN	female	1994-06-23	3	3
SA170110008	LA	male	1994-08-23	3	3
SA170110011	QY	female	1994-05-08	4	4
SA170110013	WWT	male	1995-11-17	4	4
SA170110018	ZPL	male	1995-12-02	4	4
SA170110012	ZZY	male	1993-06-26	5	5
SA170110019	WFL	female	1996-02-13	5	5
NULL	NULL	NULL	NULL	NULL	NULL

图 20: problem 18

19. 查询年龄最大和最小学的年龄差距 (只考虑年)

```
/*                                        */
select max(year(now())-year(BIRTHDAY))-min(year(now())-year(BIRTHDAY))
from student;
```

max(year(now())-year(BIRTHDAY))-min(year(...
3

图 21: problem 19

20. 查询选修"Convex_Optimization" 课程的男同学的成绩表

```
/*                        */
select * from score where NO in (select NO from student where SEX='male')
    and CNO in (select CNO from course where CNAME='Convex_Optimization');
/*                            */
select * from score a
where not exists (select * from score b where a.CNO=b.CNO and a.DEGREE<b.DEGREE);
```

NO	CNO	DEGREE
SA170110006	20181009	91
SA170110007	20181009	82
SA170110008	20181009	91
NULL	NULL	NULL

(a)only Convex_Optimization

NO	CNO	DEGREE
SA170110006	20181103	92
SA170110006	20181009	91
SA170110006	20181102	76
SA170110007	20181009	82
SA170110007	20181102	79
SA170110008	20181009	91
SA170110008	20181102	82
SA170110008	20181107	79
SA170110008	20181110	69
NULL	NULL	NULL

(b)all course

图 22: problem 20

21. 查询每门课最高分同学的学号，课程号和成绩

下面给出三种等价写法

```
/*1*/
select NO,a.CNO CNO,DEGREE from score a,
(
select CNO,max(DEGREE) x
from score
group by CNO)b
where a.DEGREE=x and b.CNO=a.CNO;
/*2*/
select * from score a where DEGREE in
                            (select max(DEGREE) from score where a.CNO=CNO);
/*3*/
select * from score a where not exists
                            (select * from score b where a.CNO=b.CNO and a.DEGREE<
                                b.DEGREE);
```

NO	CNO	DEGREE
SA170110002	20181110	95
SA170110003	20181103	92
SA170110004	20181102	95
SA170110006	20181103	92
SA170110006	20181009	91
SA170110008	20181009	91
SA170110013	20181104	89
SA170110014	20181106	83
SA170110015	20181107	91
SA170110019	20181105	96
NULL	NULL	NULL

图 23: problem 21

22. 查询成绩比该课程平均成绩高的同学的成绩表

```
select *
from score a where DEGREE>(select avg(DEGREE) from score where a.CNO=CNO);
```

NO	CNO	DEGREE
SA170110001	20181110	91
SA170110001	20181102	89
SA170110001	20181107	89
SA170110002	20181110	95
SA170110002	20181102	94
SA170110002	20181103	81
SA170110003	20181103	92
SA170110003	20181102	89
SA170110004	20181102	95
SA170110005	20181102	93
SA170110006	20181103	92
SA170110006	20181009	91
SA170110008	20181009	91
SA170110011	20181104	84
SA170110013	20181104	89
SA170110014	20181106	83
SA170110015	20181107	91
SA170110018	20181105	92
SA170110019	20181105	96
SA170110019	20181107	88
SA170110020	20181107	86
NULL	NULL	NULL

图 24: problem 22

23. 查询所有女教师和女同学的姓名、性别和生日

```
select NAME,SEX,BIRTHDAY from teacher where SEX='female'
union select NAME,SEX,BIRTHDAY from student where SEX='female';
```

NAME	SEX	BIRTHDAY
WKS	female	1988-12-23
SL	female	1969-07-28
XHF	female	1986-07-16
WKS	female	1995-09-12
ZDK	female	1994-08-12
UN	female	1994-06-23
QY	female	1994-05-08
XH	female	1996-10-09
LC	female	1995-11-30
WFL	female	1996-02-13
XY	female	1993-02-14

图 25: problem 23

24. 查询所有学生的学号、姓名、选课数、总成绩（包括选课数为 0）

```
/*sum(DEGREE)  null                                    sum(DEGREE)    avg(DEGREE)
      */
select NO,NAME,count(CNO),sum(DEGREE)
from student left outer join score using(NO) group by NO;
```

NO	NAME	count(CNO)	sum(DEGREE)
SA170110001	LY	4	341
SA170110002	WKS	3	270
SA170110003	WPY	3	263
SA170110004	HTN	2	163
SA170110005	ZDK	4	314
SA170110006	QDS	3	259
SA170110007	PJD	2	161
SA170110008	LA	4	321
SA170110009	UN	0	NULL
SA170110010	WX	1	81
SA170110011	QY	2	159
SA170110012	ZZY	1	78
SA170110013	WWT	1	89
SA170110014	LSS	1	83
SA170110015	XH	2	165
SA170110016	LC	1	77
SA170110017	TX	1	80
SA170110018	ZPL	2	176
SA170110019	WFL	2	184
SA170110020	XY	2	171

图 26: problem 24

25. 查询 Database 课程最高分的同班男同学

```
select * from student where SEX='male' and CLASS in
    (select CLASS from student where NO in
        (select NO from
            (select * from score where CNO in
                (select CNO from course where CNAME='database')) a
     where not exists (select * from score where CNO in
        (select CNO from course where CNAME='database') and DEGREE>a.DEGREE)));
```

NO	NAME	SEX	BIRTHDAY	CLASS	DEPART
SA170110004	HTN	male	1995-03-15	2	2
SA170110007	PJD	male	1994-06-14	2	2
SA170110010	WX	male	1995-02-24	2	2
SA170110017	TX	male	1996-05-16	2	2
NULL	NULL	NULL	NULL	NULL	NULL

图 27: problem 25

26. 查询 90 分以上至少有三位同学的课程信息以及其授课教师信息

```
select course.*,teacher.* from course,teacher
where teacher.NO=course.TNO and course.CNO in
        (select CNO from score where DEGREE>90 group by CNO having count(*)>2);
```

CNO	CNAME	TNO	NO	NAME	SEX	BIRTHDAY	PROF	DEPART
20181102	Database	TA80021	TA80021	WKS	female	1988-12-23	Instructor	11

图 28: problem 26

27. 查询授课教师是教授的课程信息及其最高分

```
select course.*,max(DEGREE) from course,teacher,score where course.TNO=teacher.NO
    and score.CNO=course.CNO and PROF='Professor' group by course.CNO;
```

CNO	CNAME	TNO	max(DEGREE)
20181104	Operating_System	TA80025	89

图 29: problem 27

2.4 视图

28. 建立选过 11 系老师课程的学生的成绩试图 (cs_student_score)，属性与 score 表一样，并且要求进行修改和插入操作时仍需保证该试图只有选过 11 系老师课程的学生

```
/*            */
create view cs_student_score
as select * from score where NO in(
 select NO from score where CNO in (select CNO from course where TNO in (select NO
      from teacher where DEPART=11)));
SELECT * FROM ST.cs_student_score;
/*                sql            */
create view cs_student_score
as select distinct score1.* from score score1,score score2,course,teacher
where score1.NO=score2.NO and score2.CNO=course.CNO and course.TNO=teacher.NO and
    teacher.DEPART=11;
SELECT * FROM ST.cs_student_score;

/*    */
delete from score where CNO between 20181104 and 20181105 or CNO=20181102;
SELECT * FROM ST.cs_student_score;
/*    */
insert into score(no,cno,degree) values('SA170110009','20190001',100);
SELECT * FROM ST.cs_student_score;
insert into score(no,cno,degree) values('SA170110009','20181106',100);
SELECT * FROM ST.cs_student_score;
```

SA170110001	20181110	91
SA170110001	20181102	89
SA170110001	20181107	89
SA170110001	20181103	72
SA170110002	20181110	95
SA170110002	20181102	94
SA170110002	20181103	81
SA170110003	20181103	92
SA170110003	20181102	89
SA170110003	20181110	82
SA170110004	20181102	95
SA170110004	20181103	68
SA170110005	20181102	93
SA170110005	20181009	76
SA170110005	20181107	74
SA170110005	20181103	71
SA170110006	20181103	92
SA170110006	20181009	91
SA170110006	20181102	76
SA170110007	20181009	82
SA170110007	20181102	79

SA170110008	20181009	91
SA170110008	20181102	82
SA170110008	20181107	79
SA170110008	20181110	69
SA170110010	20181104	81
SA170110011	20181104	84
SA170110011	20181106	75
SA170110012	20181104	78
SA170110013	20181104	89
SA170110014	20181106	83
SA170110015	20181107	91
SA170110015	20181105	74
SA170110016	20181106	77
SA170110017	20181105	80
SA170110018	20181105	92
SA170110018	20181107	84
SA170110019	20181105	96
SA170110019	20181107	88
SA170110020	20181107	86
SA170110020	20181105	85

图 30: problem 28 view

此时与 score 表数据相同!

NO	CNO	DEGREE
SA170110011	20181106	75
SA170110014	20181106	83
SA170110016	20181106	77

(a)delete

NO	CNO	DEGREE
SA170110011	20181106	75
SA170110014	20181106	83
SA170110016	20181106	77

(b)insert(teacher's dept != 11)

NO	CNO	DEGREE
SA170110009	20181106	100
SA170110009	20190001	100
SA170110011	20181106	75
SA170110014	20181106	83
SA170110016	20181106	77

(c)insert(teacher's dept = 11)

图 31: problem 28 insert and delete

29. 删除视图 cs_student_score

```
drop view cs_student_score;
/*                view        */
show table status where comment='view';
```

Name	Engine	Version	Row_format	Rows	Avg_row_length	Data_length	Max_data_length

图 32: problem 29

为空说明没有视图（已删除）！

2.5 触发器

30. 为数据库创建书法器，目标：维持学生数量更新的一致性

a) 创建关系表：系别 Dept(DNO,s_count) 其中 DNO 是系编号 (int) 非空，主键；s_count 是每个系的学生人数；

```
create table Dept
(
DNO int primary key not null,
s_count int
);
SHOW COLUMNS FROM Dept;
```

Field	Type	Null	Key	Default	Extra
DNO	int(11)	NO	PRI	NULL	
s_count	int(11)	YES		NULL	

图 33: problem 30(a)

b) 根据 student 关系表，补全 Dept 中 s_count 的数据

```
insert into Dept select DEPART,count(*) from student group by DEPART;
SELECT * FROM ST.Dept;
```

DNO	s_count
1	7
2	6
3	2
4	3
5	2
NULL	NULL

图 34: problem 30(b)

c) 为关系表 student 创建两个触发器，使得每插入 (删除) 一条学生记录，Dept 表中对应系的人数加 1(减 1)

```
create trigger s_count_up after insert on student
for each row update Dept set s_count=s_count+1 where DNO=new.CLASS;

create trigger s_count_down after delete on student
for each row update Dept set s_count=s_count-1 where DNO=old.CLASS;
```

Name	Event	Timing
s_count_up	INSERT	AFTER
s_count_down	DELETE	AFTER

图 35: problem 30(c)

d) 检验触发器是否工作

```sql
/*    */
insert into student(no,class) values('test1',1),('test2',2),('test3',3);
SELECT * FROM ST.Dept;
/*    */
delete from student where NO='test1';
delete from student where NO='test2';
delete from student where NO='test3';
delete from student where DEPART=5;
SELECT * FROM ST.Dept;
```

DNO	s_count
1	8
2	7
3	3
4	3
5	2
NULL	NULL

(a) 插入

DNO	s_count
1	7
2	6
3	2
4	3
5	0
NULL	NULL

(b) 删除

图 36: problem 30(d)